

Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title:

AI Study planner

Team Name:

Smart Workers

Team Members:

1.G. Sai Varshini

2.P. Lahari

3.G. Gayathri

4.K. Naga Sujatha

5.K. Deepika

Phase-1: Brainstorming & Ideation

Objective:

Creating an AI based study planner.

The purpose of the AI-based Study Plan project is to create personalized, adaptive study schedules that enhance learning efficiency and academic performance using artificial intelligence

An AI study planner enhances learning by creating personalized, efficient schedules tailored to individual needs, improving time management, retention, and productivity. It reduces stress, provides data-driven insights, and makes education more accessible and inclusive.

Key Points:

Problem Statement: The problem statement for an AI study planner is: "Students and learners often struggle with inefficient time management, lack of personalized study plans, and difficulty tracking progress, leading to stress and suboptimal learning outcomes. An AI study planner aims to address these challenges by providing adaptive, data-driven scheduling and resource recommendations tailored to individual needs."

Proposed Solution: The proposed solution is: "An AI-powered study planner that creates personalized, adaptive schedules based on individual learning patterns, tracks progress, and provides smart recommendations for resources and study techniques. It integrates spaced repetition, deadline reminders, and performance analytics to optimize learning efficiency and reduce stress."

Target Users: The target users are **students, professionals, and lifelong learners** seeking structured and efficient study plans. Primary beneficiaries include **academic students, exam aspirants, and individuals pursuing skill development or certifications**, as the AI study planner enhances productivity, retention, and time management.

Expected Outcome: The expected outcome of the AI study planner project is: **Improved learning efficiency, better time management, and enhanced academic or professional performance for users.** Additionally, it aims to reduce stress, increase motivation, and foster a habit of organized, goal-oriented learning through personalized, data-driven study plans.

Phase-2: Requirement Analysis

Objective:

Technical requirements:

1. **Languages**: Python (for AI/ML), JavaScript (for front-end), and SQL (for database management).
2. **Frameworks**: TensorFlow/ PyTorch (AI/ML), React.js/Flutter (front-end), and Django/Node.js (back-end).
3. **Tools**: Git (version control), Docker (containerization), Tableau/Power BI (analytics), and AWS/Google Cloud (cloud infrastructure).

Functional requirements:

1. **Personalized Scheduling**: Adaptive study plans based on user goals, deadlines, and learning patterns.
2. **Progress Tracking**: Real-time monitoring of study hours, task completion, and performance metrics with visual dashboards.
3. **Smart Recommendations**: AI-driven resource suggestions (e.g., videos, quizzes) and spaced repetition for improved retention.

Key Points:

1. **Technical Requirements**: (Languages, frameworks, tools)
 2. **Functional Requirements**: (Features the project must have)
 3. **Constraints & Challenges**: (Any limitations or risks)
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Phase-3: Project Design

Objective:

- Create the architecture and user flow.

Key Points:

1. **System Architecture Diagram**: (Simple sketch or flowchart)
 2. **User Flow**: (How a user will interact with the project)
 3. **UI/UX Considerations**: (If applicable, wireframe or basic layout)
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Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down the tasks using Agile methodologies.

Key Points:

1. **Sprint Planning:** (Divide work into tasks for each team member)
 2. **Task Allocation:** (Who will do what?)
 3. **Timeline & Milestones:** (Set short deadlines for each task)
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Phase-5: Project Development

Objective:

- Code the project and integrate components.

Key Points:

1. **Technology Stack Used:** (List of programming languages, APIs, etc.)
 2. **Development Process:** (Steps followed for coding)
 3. **Challenges & Fixes:** (Mention any obstacles faced and how they were solved)
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Phase-6: Functional & Performance Testing

Objective:

- Ensure the project works as expected.

Key Points:

1. **Test Cases Executed:** (List the scenarios tested)
 2. **Bug Fixes & Improvements:** (Mention fixes made)
 3. **Final Validation:** (Does the project meet the initial requirements?)
 4. **Deployment (if applicable):** (Hosting details or final demo link)
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Final Submission

1. **Project Report Based on the templates**
 2. **Demo Video (3-5 Minutes)**
 3. **GitHub/Code Repository Link**
 4. **Presentation**
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